STALKS OF THE PUSHFORWARD ALONG A FINITE MORPHISM AND VANISHING OF ITS HIGHER DIRECT IMAGES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for both parts and for both settings of part (1).

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Finite morphism of schemes and its fibres. Same chapter together with the chapter Morphisms, the fact that the fibre of a finite morphism over a geometric point is finite and consists of geometric points. Supplies the finite index set over which the product and the direct sum are taken.
- Stalks of a pushforward and exactness of the pushforward along a finite morphism. Same chapter, the description of the stalk of f_*F at a geometric point and the exactness properties used to conclude the vanishing of the higher direct images. Supplies the computation in part (1) and the vanishing in part (2).

Source reference. Original theorem 03Q9: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8015–8015. Navigation pointer invoked at proof line 8269; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03QJ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4186–4186. Navigation pointer invoked at proof line 8277; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 03QO: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8207–8207. Navigation pointer invoked at proof line 8284; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original theorem 03PU: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3546–3546. Navigation pointer invoked at proof line 8286; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Proposition 0.1. *Let $f : X \to Y$ be a finite morphism of schemes.*

(1) *For any geometric point $\overline{y} : \mathrm{Spec}(k) \to Y$ we have*

$$(f_*\mathcal{F})_{\overline{y}} = \prod\nolimits_{\overline{x}:\mathrm{Spec}(k)\to X,\ f(\overline{x})=\overline{y}} \mathcal{F}_{\overline{x}}.$$

for $\mathcal{F}$ in $Sh(X_{étale})$ and

$$(f_*\mathcal{F})_{\overline{y}} = \bigoplus\nolimits_{\overline{x}:\mathrm{Spec}(k)\to X,\ f(\overline{x})=\overline{y}} \mathcal{F}_{\overline{x}}.$$

for $\mathcal{F}$ in $Ab(X_{étale})$.

(2) *For any $q \geq 1$ we have $R^q f_*\mathcal{F} = 0$ for $\mathcal{F}$ in $Ab(X_{étale})$.*

Proof. Let $X^{sh}_{\overline{y}}$ denote the fiber product $X \times_Y \mathrm{Spec}(\mathcal{O}^{sh}_{Y,\overline{y}})$. By Theorem 03Q9 the stalk of $R^q f_*\mathcal{F}$ at $\overline{y}$ is computed by $H^q_{étale}(X^{sh}_{\overline{y}}, \mathcal{F})$. Since f is finite, $X^{sh}_{\overline{y}}$ is finite over $\mathrm{Spec}(\mathcal{O}^{sh}_{Y,\overline{y}})$, thus $X^{sh}_{\overline{y}} = \mathrm{Spec}(A)$ for some ring A finite over $\mathcal{O}^{sh}_{Y,\overline{y}}$. Since the latter is strictly henselian, Lemma 03QJ implies that A is a finite product of henselian local rings $A = A_1 \times \ldots \times A_r$. Since the residue field of $\mathcal{O}^{sh}_{Y,\overline{y}}$ is separably closed the same is true for each A_i. Hence A_i is strictly henselian. This implies that $X^{sh}_{\overline{y}} = \coprod_{i=1}^r \mathrm{Spec}(A_i)$. The vanishing of Lemma 03QO implies that $(R^q f_*\mathcal{F})_{\overline{y}} = 0$ for $q > 0$ which implies (2) by Theorem 03PU. Part (1) follows from the corresponding statement of Lemma 03QO. □